



EEE 6778: Applied Machine Learning II

Course Overview

Professor:

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Office Hours:

Thursdays 11:45-1:45pm

MALA4112

Contact:

(1) Slack, (2) Email: aramirezsalgado@ufl.edu

Topics:

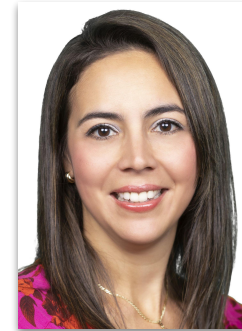
AI+IoT (AloT), edge AI, intelligent embedded systems

Research:

Teaching and learning approaches to enhance the representation of all populations in computer engineering

Previous work:

Undergrad and grad courses in CS, ECE, and curriculum and instruction +
Research in inclusive and equitable learning environments + Civil engineering company

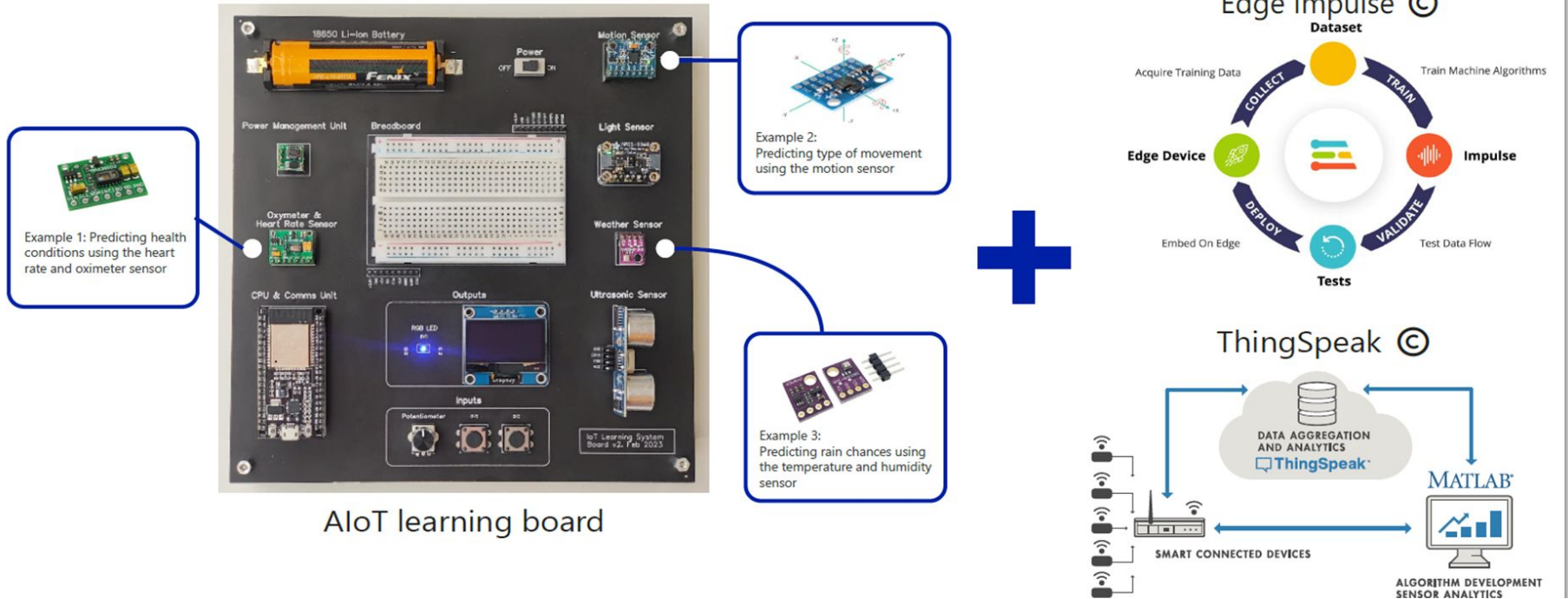


About me

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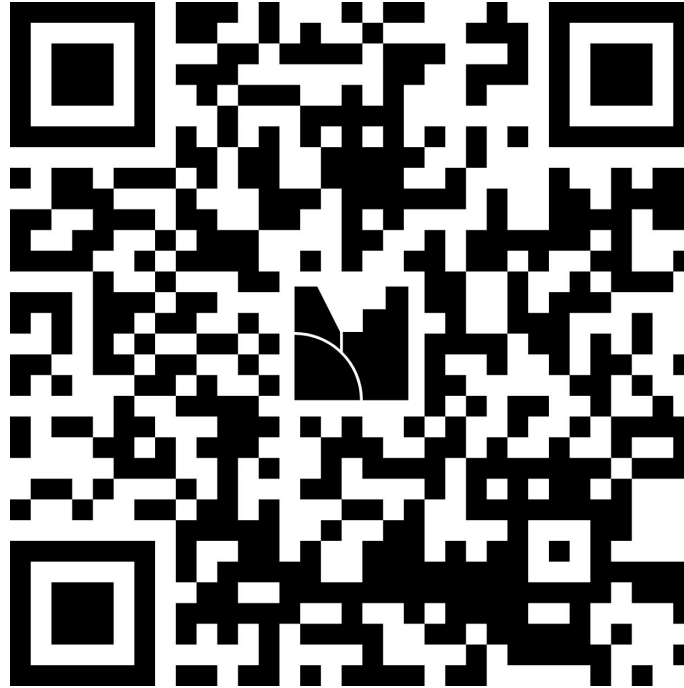
Curricular Interest – Edge AI and AIoT - Core Member of the [Nelms Institute for the Connected World](#)

About me

- **What did you do over the summer break?**
 - Is there anything cool that you did that you want to share with us?



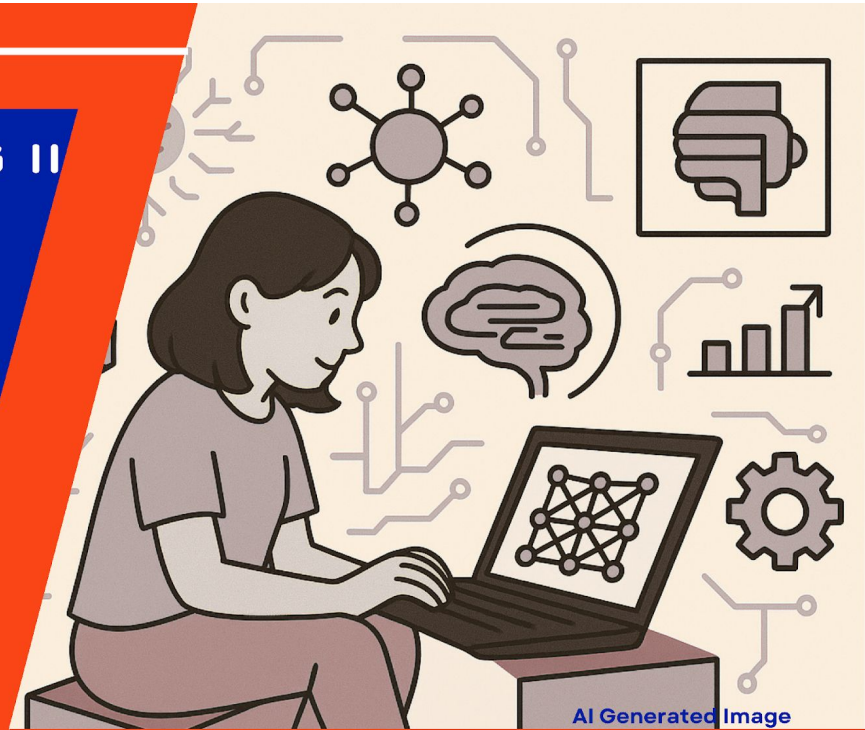
Things I want to know about you



Your expectations

APPLIED MACHINE LEARNING II

This course offers an advanced, application-oriented study of machine learning and deep learning. Students will build both theoretical understanding and practical expertise through hands-on labs and projects. Topics include graphical models, computer vision, recursive neural networks, transformer architectures, and edge AI, to solve real-world problems.



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Upon completion of this course, students will be able to:

- Design and implement neural networks using foundational principles of machine learning and **optimization**.
- Apply **probabilistic graphical models**, including directed models (e.g., Bayesian networks) and undirected models (e.g., Markov Random Fields), to represent and reason under uncertainty in real-world systems.
- Fine-tune and evaluate deep learning architectures, including **CNNs**, **Autoencoders**, and **GANs**, for tasks such as image classification, compression, and generation.
- Leverage sequence modeling techniques, such as **RNNs**, **LSTMs**, and **Transformers**, to analyze time series and natural language data, selecting appropriate architectures based on problem constraints.
- Deploy and optimize deep learning models on **edge devices**, using techniques like pruning, quantization, and distillation to meet performance, power, and memory requirements in real-time applications.
- Integrate advanced architectures, including **Graph Neural Networks (GNNs)**, multimodal systems (e.g., CLIP), and diffusion models, to design innovative solutions to open-ended problems across domains.
- Effectively **communicate technical insights** and project outcomes, presenting design decisions, model performance, and implications to both technical and non-technical stakeholders through written reports and oral presentations.

Course objectives

Interactive mini-lecture

In-class labs and
peer feedback

Project!

- **Canvas.**

- Canvas is our principal hub
- Here I will post the resources, assignments and deadlines
- All deadlines will be on Sundays at 11:59pm (one exception detailed later)
- Inbox

The screenshot shows the Canvas LMS interface for the course 'APPLIED MACHINE LEARNING II' (EEE6778) in the Engineering Education Department. The left sidebar contains a navigation menu with items like Home, Syllabus, Announcements, Modules, Assignments, Grades, Course Reserves, People, Zoom Conferences, Instructor Tools, LockDown Browser, Discussions, Pages, Files, Outcomes, Rubrics, Quizzes, Collaborations, Course Analytics, GatorEvals, and Settings. The main content area displays the course title, a description of the course, and a list of modules. The modules list includes: General Information and Resources, Module 1. Introduction to Deep Learning and Neural Networks, Module 2. Convolutional Neural Networks (CNNs), Module 3. Autoencoders and Generative Adversarial Networks (GANs), Module 4. Recurrent Neural Networks (RNNs), Module 5. Transformers, and Module 6. Advanced Deep Learning Architectures. The interface also features a top navigation bar with 'Assign To' and 'Edit' buttons, and a bottom navigation bar with 'Welcome', 'Schedule', and 'Modules' tabs.

Course structure

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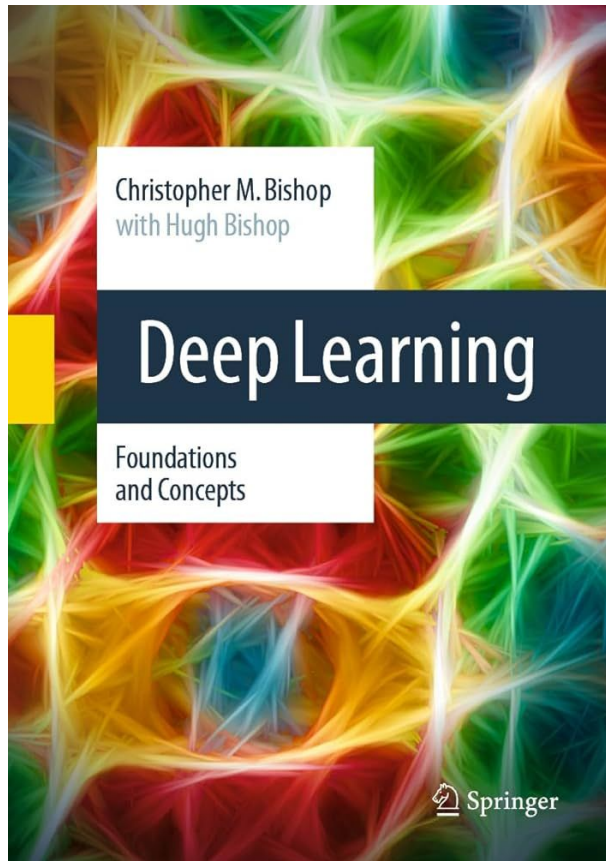


- **GitHub.**
 - GitHub is our secondary hub
 - Here I will post the labs and project repositories



- **Slack.** [uf-eee6778-AppliedML2-Fall25](#)
 - This resource is optional and intended to supplement office hours and student interaction
 - No official communication/submission happens over Slack

Course structure



- **Main Textbook**
 - **Deep Learning: Foundations and Concepts.** Christopher M. Bishop and Hugh Bishop. Springer, 2023
 - Available as electronic copy with Course Reserves (in Canvas)
 - All materials and slides will be available in Canvas
- **Software**
 - **PyTorch** will be the core deep learning framework
 - **HiPerGator** TBD

Materials and Software

Week	Subject	Assignments
1	Course Introduction and Introduction to Advanced Machine Learning and Deep Learning	
2	Optimization Techniques	Project Proposal (5%)
3	Convolutional Neural Networks (CNNs)	
4	Probabilistic Graphical Models: Directed and Undirected	Elevator Pitch (10%) Elevator Pitch Peer-Feedback (5%)
5	Generative Adversarial Networks (GANs)	
6	Autoencoders	Project Delivery 1 (5%)
7	Recurrent Neural Networks (RNNs)	
8	Transformers	Project Delivery 2 (5%)
9	Transformers	Project Peer-Feedback (5%)
10	Advanced Generative Models and Multimodal Architectures	
11	Advanced Generative Models and Multimodal Architectures	Project Delivery 3 (10%)
12	Model Compression and Edge AI	
13	Graph Neural Networks (GNNs)	Project Delivery 4 (5%)
14	Modern Approaches in Recommendation Systems	Project Final Code Repository (10%)
Thanksgiving		
15	Final Project Reports and Presentations	Project Final Report (5%) Final Presentation (10%) Final Presentation Peer Feedback (5%)

Course schedule



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Assignment	Total Points	Percentage of Final Grade
Project deliverables	Specified in the table above	45%
Labs		20%
Presentations		20%
Peer Feedback		15%
		100%

Project Deliverables (45%)

An open-ended semester-long project where you INDIVIDUALLY will develop a solution or exploration of your choice.

Labs (20%)

A cumulative set of required labs that build technical foundations. This playground ensures that you gain hands-on experience with the core models covered in class. To maintain the pace of the course, each lab has its own deadline aligned with project milestones, and late submissions follow the course late policy.

Grading

Presentations (20%)

Elevator Pitch – 10%

Final Presentation – 10%

Peer Feedback (15%)

Peer feedback is tied directly to in-class activities. A portion of the grade is based on **mini reports** submitted during class sessions

Elevator Pitch Peer Feedback – 5%

In-class mini report: 3%

Quality of rubric-based comments: 2%

Project Peer Feedback – 5%

In-class mini report: 3%

Quality of rubric-based comments: 2%

Final Presentation Peer Feedback – 5%

In-class mini report: 3%

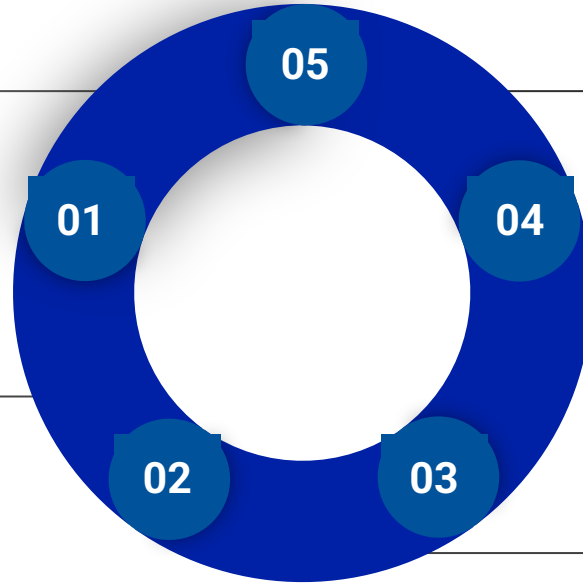
Quality of rubric-based comments: 2%

Assignment	Total Points	Percentage of Final Grade
Project deliverables	Specified in the table above	45%
Labs		20%
Presentations		20%
Peer Feedback		15%
		100%

Grading

**Interdisciplinary
skill
development.**

**Project-based
learning**



**Cutting-edge AI
expertise**

**Real-world
application**

**Industry
relevance**

Why is this important



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- I will prepare course lectures with the expectation that **students will attend class synchronously** and bring a computer to follow along with any practical implementations.
- All assignments are due **on time**; work up to 24 hours late loses 5%, up to 72 hours late loses 10%, and after that no credit is given. Peer feedback and presentations must be completed in class on the scheduled date and cannot be made up.
- Unless otherwise noted, there are **no group or collaborative assignments** in this course.
- If you're unsure about the acceptable use of AI tools for your work, **ask for clarification!**
- Students will complete this course with honor and integrity, or not at all. Submissions which are believed to be not entirely a student's own work will be **reported to administration** for disciplinary action. In ALL cases, I will recommend the following sanctions be imposed on that student or students:

A failing grade (an 'E') for the course

That you not be allowed to drop the course for any reason



**Read the
syllabus!**

Course Policies



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✓ As your professor, I will:

- Come to class prepared
- Strive to use class time wisely by structuring relevant and engaging activities
- Avoid busywork by relating assignments to our classroom sessions & practical applications of concepts
- Take into account student feedback
- Recognize that learning is a partnership between instructor & student

✓ As the student, you should:

- Recognize that learning is a partnership between instructor & student
- Acknowledge the difference between being a “student” and being a learner
- Engage in activities & be “present” in class
- Be respectful to everyone in our community
- Communicate with me immediately should any problems arise
- Appreciate the important role you play in how valuable any course is to you

Tips to be successful

AI Education and Training

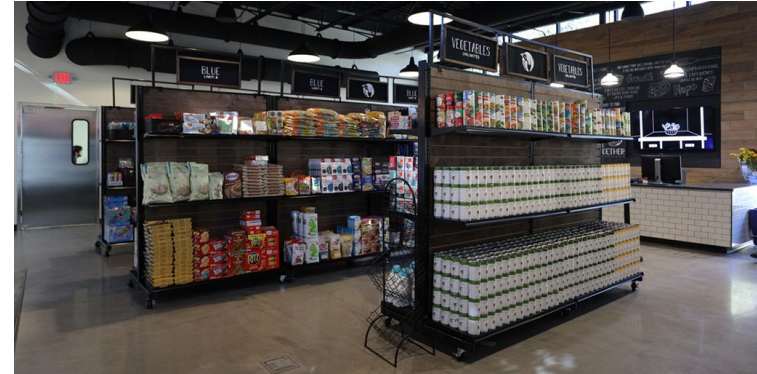
<https://docs.rc.ufl.edu/training/ai/>

Depot Park

<https://www.depotpark.org/>

Lake Wauburg

<https://recsports.ufl.edu/locations/lake-wauburg/>



Off the shelf - UF Resources

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Any Questions?

Let's get started!

Before we move forward please complete the Warm-up Quiz included in Module 1 in Canvas

This quiz is ungraded and is solely intended to help me assess your prior knowledge.

Let's get started!

Module 1 in Canvas has two cheat sheets to cover linear algebra and probability foundations

Linear Algebra Cheat Sheet for Machine and Deep Learning

Andrea Ramirez-Salgado

Introduction

Linear algebra is the foundation of many machine learning and deep learning algorithms. Understanding how data is represented and manipulated mathematically is essential for building efficient and accurate models. This cheat sheet summarizes key linear algebra concepts relevant to these fields. This cheat sheet summarizes key linear algebra concepts relevant to these fields based on Chapter 2 of the Deep Learning textbook (www.deeplearningbook.org) by Goodfellow et al. (2016).

1. Scalars, Vectors, Matrices, and Tensors

- **Scalar:** A single numerical value, often used for weights, biases, or loss values in machine learning models. Denoted by lowercase italics (e.g., a).
- **Vector:** A one-dimensional array of numbers, used to represent features, weights, or data points. Denoted by bold lowercase (e.g., $\mathbf{v}$).
- **Matrix:** A two-dimensional array of numbers, commonly used for storing datasets, transformation operations, or model parameters. Denoted by bold uppercase (e.g., $\mathbf{M}$).
- **Tensor:** A generalization of vectors and matrices to higher dimensions, essential for handling multi-dimensional data in deep learning (e.g., images, video, text sequences).

Probability and Information Theory Cheat Sheet for Machine and Deep Learning

Andrea Ramirez-Salgado

Probability theory and information theory are foundational to understanding uncertainty and quantifying information in machine and deep learning models. This cheat sheet summarizes key linear algebra concepts relevant to these fields based on Chapter 3 of the Deep Learning textbook (www.deeplearningbook.org) by Goodfellow et al. (2016).

1. Why Probability?

Machine learning systems must handle uncertainty due to:

- **Inherent Stochasticity:** Random behavior in systems (e.g., quantum mechanics).
- **Incomplete Observability:** Lack of access to all variables (e.g., Monty Hall problem).
- **Incomplete Modeling:** Simplified models lead to uncertainty (e.g., discretizing continuous data).

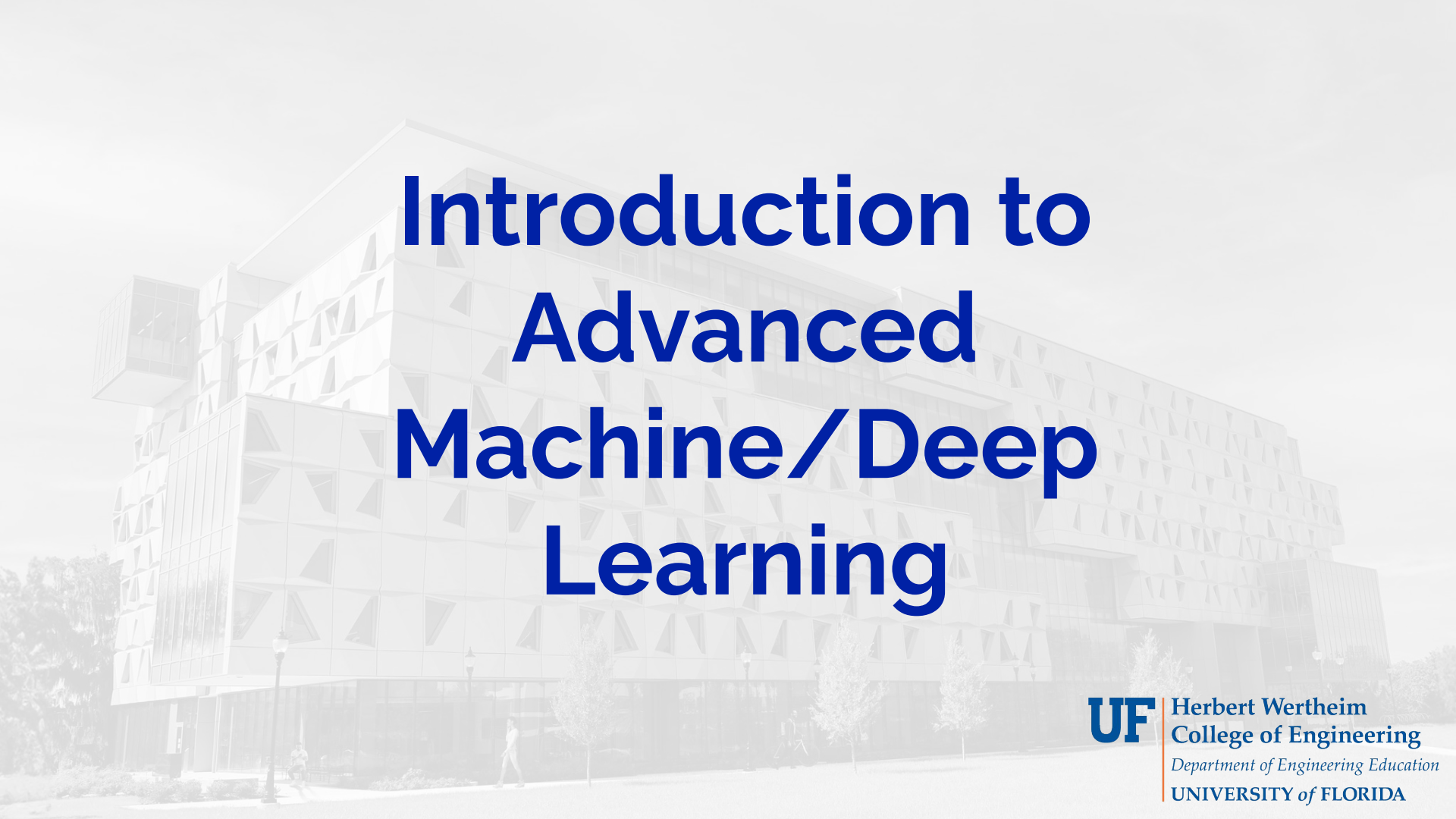
2. Types of Probability

- **Frequentist Probability:** Based on event frequencies in repeated experiments.
- **Bayesian Probability:** Represents degrees of belief based on prior knowledge.

Let's get started!

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Introduction to Advanced Machine/Deep Learning



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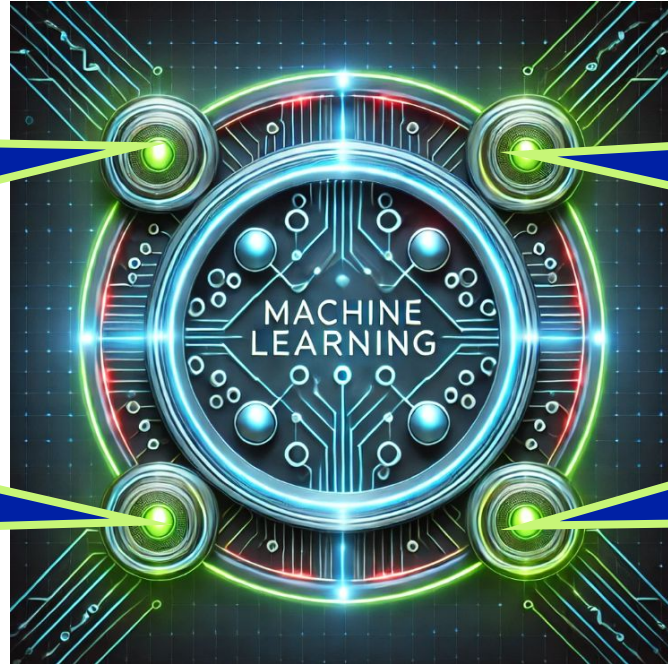
Machine Learning

Unsupervised
Learning
models

Supervised
Learning
models

Reinforcement
Learning

Deep
Learning



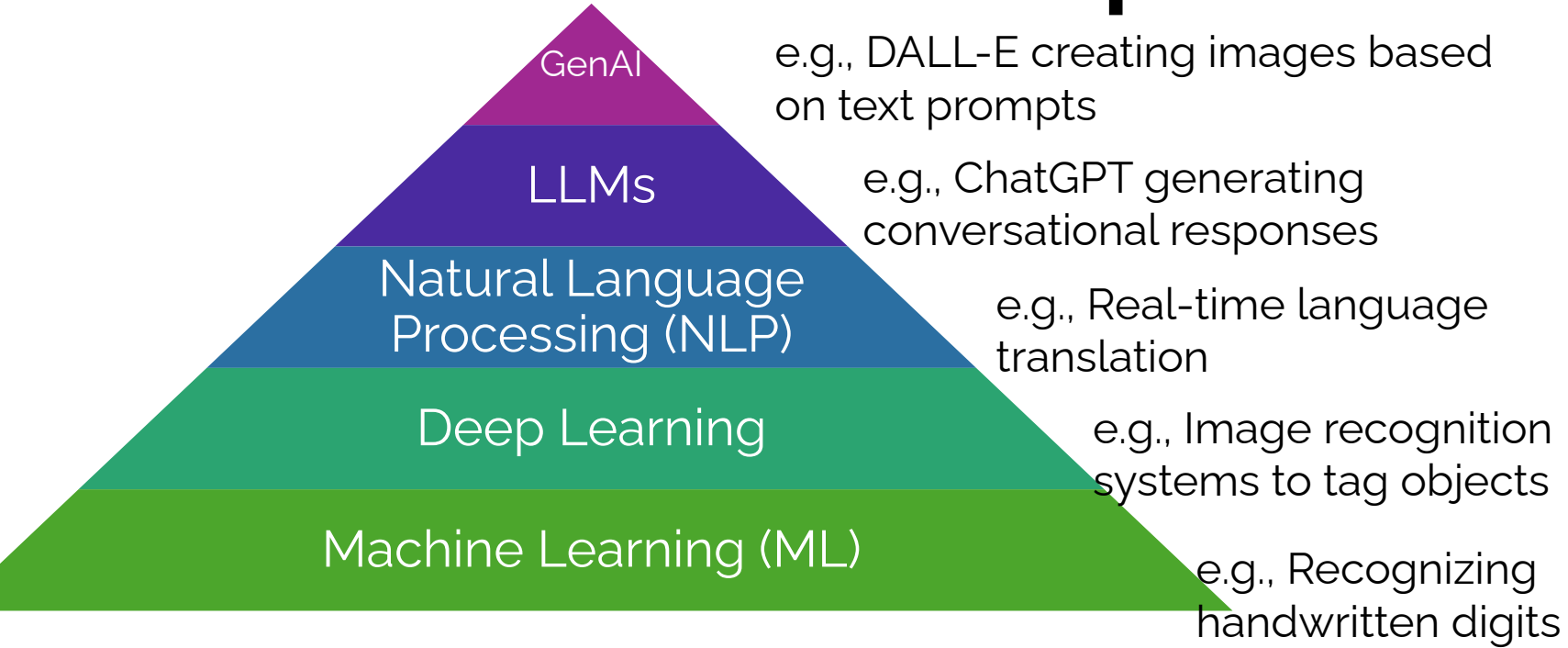
AI-generated image using DALL-E. Prompt: A futuristic circle with four external nodes in the edge and the label "machine learning" in the inner circle

AI Overview

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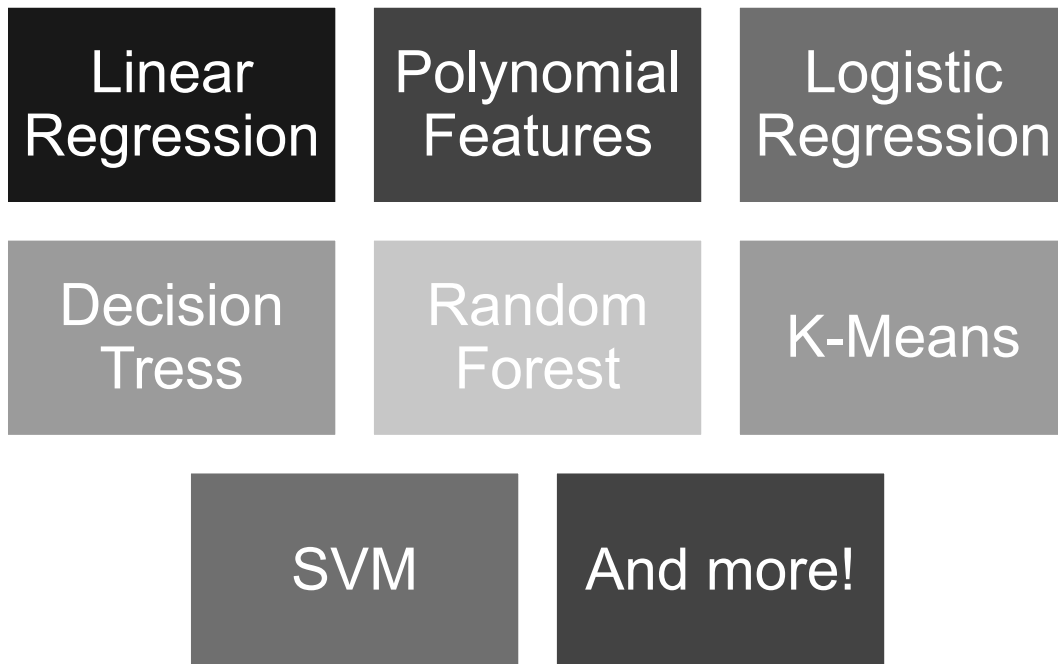
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AI Branches amplified



AI Overview

What models you remember?



Traditional Machine Learning

Limitations of
traditional ML: manual
feature engineering,
scalability



Deep learning automates
feature extraction and
probabilistic models
capture causal
dependencies!



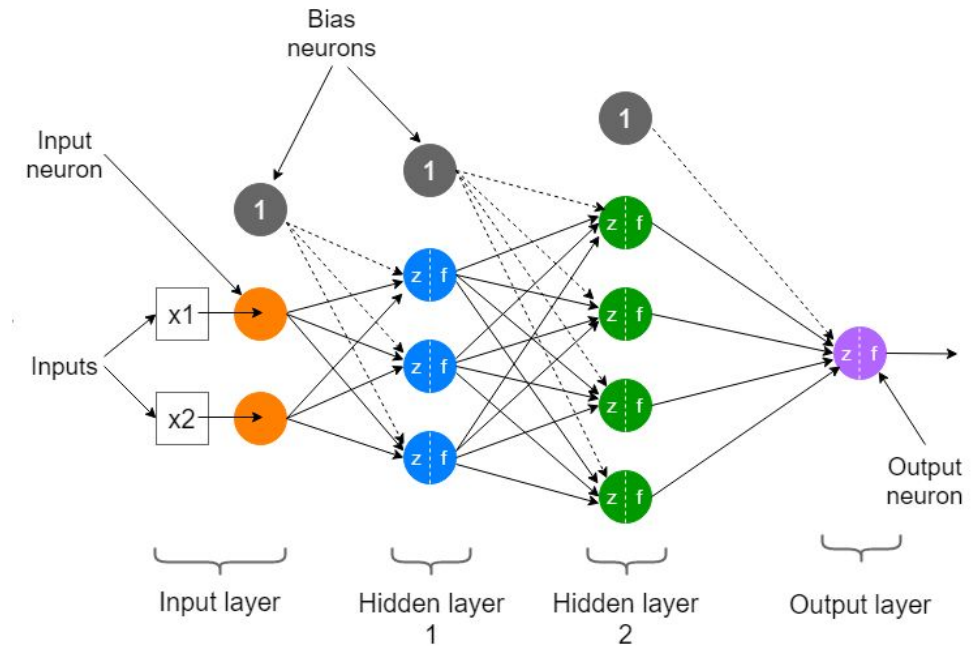
Success in image
recognition, NLP, and
autonomous systems

Motivation for Advancement



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- **Neural networks:** neurons, layers, activation functions
- Shallow vs. deep models
- **Universal Approximation Theorem:** A neural network can learn any pattern or function if it has enough hidden neurons and uses a non-linear activation.



Deep Learning Foundations



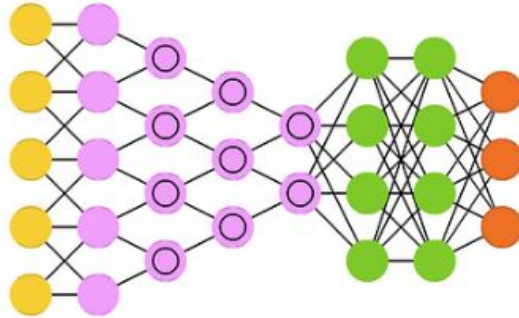
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- Feedforward Neural Networks (FNNs)
- Convolutional Neural Networks (CNNs)
- Recurrent Neural Networks (RNNs) & LSTMs
- ...

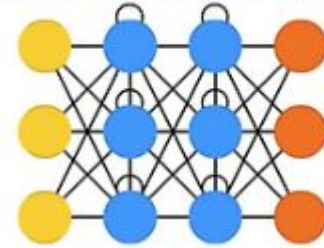
Feed Forward (FF)



Deep Convolutional Network (DCN)



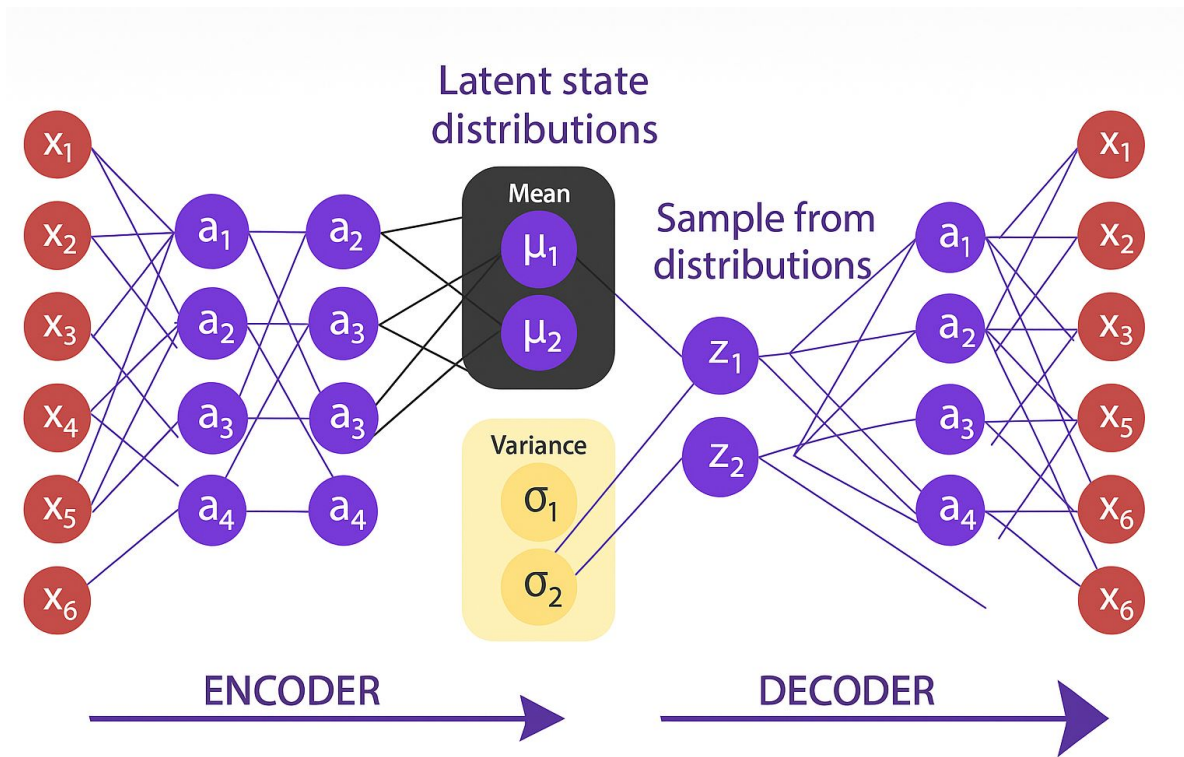
Recurrent Neural Network (RNN)



Key Deep Learning Architectures



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Transformers and multimodal AI

Probabilistic Deep Learning



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- **CNNs**: Image recognition and computer vision tasks like object detection.
- **Autoencoders**: Dimensionality reduction and anomaly detection.
- **GANs**: Generating realistic synthetic images and data.
- **RNNs**: Sequence modeling for time series and speech recognition.
- **NLP**: Understanding and processing human language for tasks like sentiment analysis.
- **Transformers and LLMs**: Advanced language understanding and generation (e.g., ChatGPT).
- **Edge AI**: Real-time inference on devices for applications like autonomous vehicles and IoT.

Going back to our class

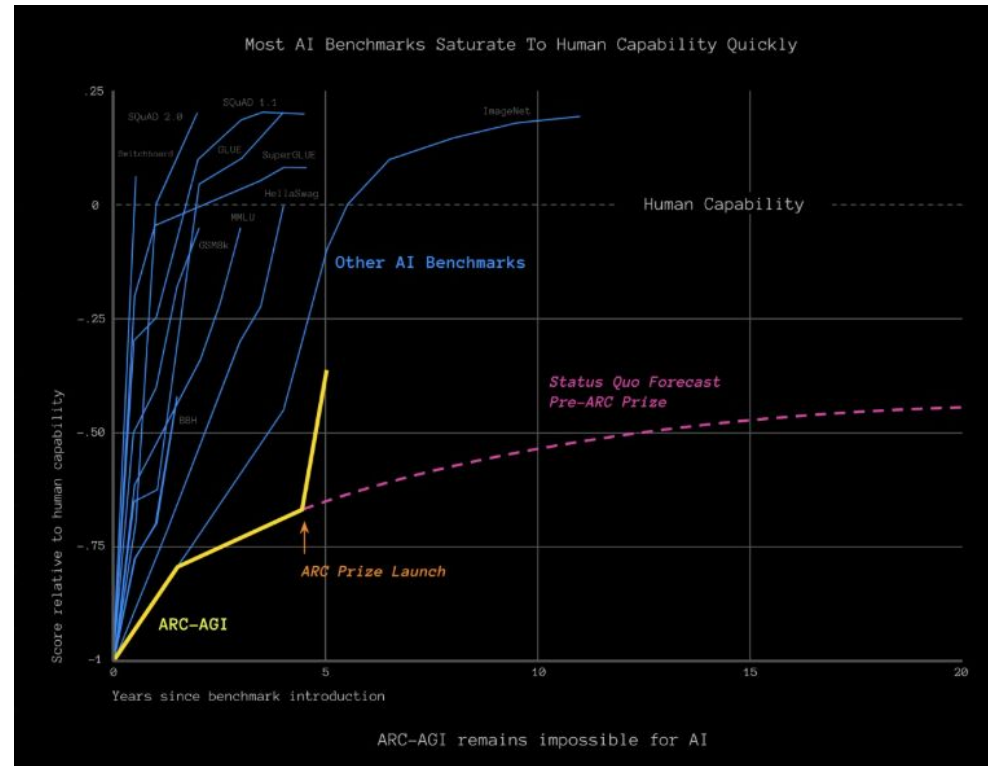
Whats next? [Watch François Chollet talking about OpenAI o-models and ARC/AGI](#) (Abstraction and Reasoning Corpus/Artificial General Intelligence

- ARC 2025 Prize

ARC-AGI

Most AI benchmarks measure skill. But skill is not intelligence. General intelligence is the ability to efficiently acquire new skills. Chollet's unbeaten 2019 Abstraction and Reasoning Corpus for Artificial General Intelligence (ARC-AGI) is the only formal benchmark of AGI progress.

It's easy for humans, but hard for AI.



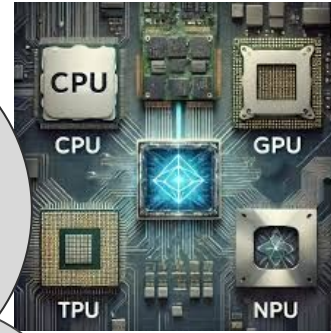
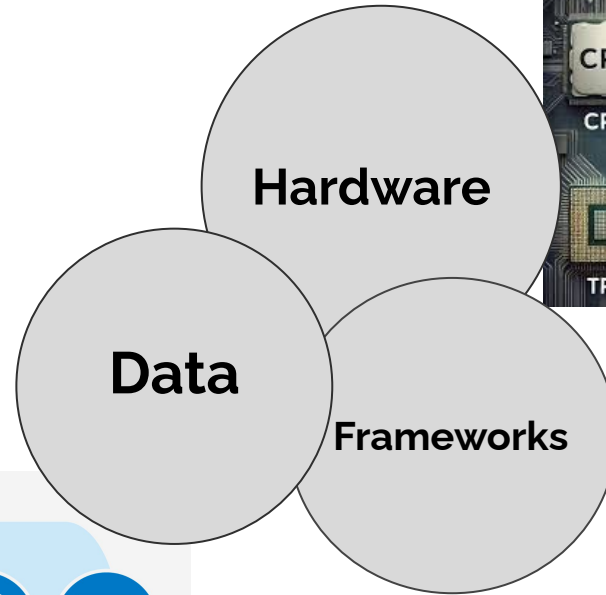
What is coming next?



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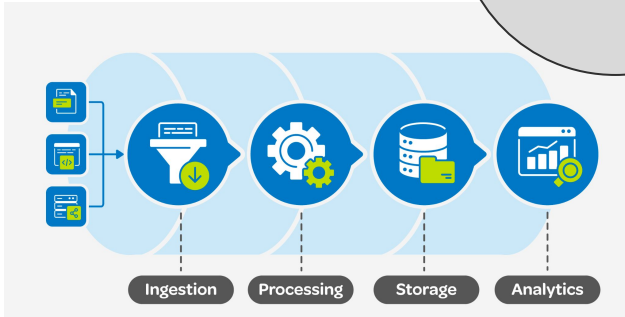
- **Many technical challenges involved!**
- Forward and backward propagation
- Gradient descent and optimizers (SGD, Adam)
- Overfitting and regularization (dropout, batch norm)

Training Advanced Networks



TensorFlow

 PyTorch



Deep Learning Ecosystem



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Applications

Healthcare, finance, autonomous systems, NLP, computer vision, entertainment and media, industry 4.0, e-commerce, agriculture, cybersecurity, IoT, etc.

Challenges

Interpretability, data needs, ethics, computational demands, maintenance and deployment, security, scalability, energy consumption, etc.

Real-World Application and Challenges



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